Practical 7

System has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

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GNU nano 7.2

```c
#include <stdio.h>

#define P 5
#define R 4

int main() {
    int alloc[P][R], max[P][R], avail[R];
    int need[P][R];
    int finish[P] = {0}, safeSeq[P];
    int i, j, k = 0;

    printf("Enter Allocation Matrix:\n");
    for(i=0;i<P;i++)
        for(j=0;j<R;j++)
            scanf("%d",&alloc[i][j]);

    printf("Enter Maximum Matrix:\n");
    for(i=0;i<P;i++)
        for(j=0;j<R;j++)
            scanf("%d",&max[i][j]);

    printf("Enter Available Resources:\n");
    for(i=0;i<R;i++)
        scanf("%d",&avail[i]);

    for(i=0;i<P;i++)
        for(j=0;j<R;j++)
            need[i][j] = max[i][j] - alloc[i][j];

    int count = 0;

    while(count < P) {
        int found = 0;

        for(i=0;i<P;i++) {
            if(finish[i] == 0) {
                int flag = 0;

                for(j=0;j<R;j++) {
                    if(need[i][j] > avail[j]) {
                        flag = 1;
                        break;
                    }
                }

                if(flag == 0) {
                    for(j=0;j<R;j++)
                        avail[j] += alloc[i][j];

                    safeSeq[k++] = i;
                    finish[i] = 1;
                    found = 1;
                    count++;
                }
            }
        }

        if(found == 0) {
            printf("\nSystem is NOT in safe state\n");
            return 0;
        }
    }

    printf("\nSystem is in SAFE state\nSafe Sequence: ");
    for(i=0;i<P;i++)
        printf("P%d ", safeSeq[i]);

    return 0;
}
```

Output:

```
Enter Allocation Matrix:
0 1 0 3
2 0 0 1
3 0 2 1
2 1 1 0
0 0 2 0
Enter Maximum Matrix:
7 5 3 4
3 2 2 2
9 0 2 2
2 2 2 2
4 3 3 3
Enter Available Resources:
3 3 2 2

System is in SAFE state
Safe Sequence: P1 P3 P4 P0 P2
```